

Query Expansion and Key Specialization in Transformer Attention Geometry

Abstract

The projection of queries and keys are central to the attention mechanism in Transformer architectures. While they are mathematically symmetric, they play different roles in attention mechanisms. The question of whether there is an effect from their functional distinction on their geometric development in training remains unanswered. We investigate the problem through the training of small GPT-like Transformers on character-level WikiText-103 for three different depths (4, 6, and 8 layers), three types of initialization for queries and keys, and four random seeds, resulting in 36 runs and 54 trajectories of average layers across seeds. We track the effective dimensionality of those layers using participation ratios and discover that effective dimension of queries expand while keys shrink, and that $PR_Q - PR_K$ is positive in all trajectories studied. In connection to attention, the shrinking of keys leads to a narrower spectrum of $QK^\top$ and more peaked attention weights. In order to determine if this connection is causal or coincidental, we directly control the spectrum of keys during training across five seeds: restricting it to make it shrink sharpens the attention with high directional confidence, while keeping it constant to the level of initial dispersion makes attention softer. Additional token-level checkpoint analyses show that the monotonic paired-contrast trend is not universal across pretrained families, but survives as an early-training regime that later decays over a full pretraining run, and the link between interaction-rank geometry and attention entropy remains visible in several models.

Keywords: Transformers Attention, Mechanistic Interpretability, Query-Key Geometry, Attention Entropy, Geometric Representation Learning

1. Introduction

The self-attention mechanism in Transformer-based models is computed using the compatibility matrix $QK^\top$. Queries and keys are often described as symmetric components of the attention mechanism Vaswani et al. (2017). In spite of their mathematical symmetry, the two matrices serve different functional roles: queries represent the information sought by each token position, whereas keys represent the information made available for retrieval. Meanwhile, initialization in deep neural networks is typically viewed as an optimization detail, with Gaussian and Xavier-based schemes primarily used to ensure stable training before the effects of gradient descent dominate Glorot and Bengio (2010). However, because attention depends directly on the interaction between W_Q and W_K , the initialization of these matrices may influence the formation of attention patterns during training. Consequently, even when models achieve comparable validation performance, different initialization schemes may lead to distinct attention geometries.

This leads to the main question posed in the paper: do query and key projections follow the same evolution path in terms of effective dimensionality during the training process, or is there a different evolution pattern followed by each one? The answer is sought via participation ratios, spectral-sensitive effective dimensionality measures, wherein PR_Q

captures how much of the spectral space is being used by query representations, and PR_K for key representations

The central result is that there are systematic differences between the geometric paths of query and key projections. The participation of queries increases strongly with respect to depth, seeds, and initialization techniques. The participation of keys diminishes most prominently in the Independent Xavier and Disjoint Support Xavier initializations of W_Q and W_K . The contrast between the two slopes, $\text{PR}_Q - \text{PR}_K$, is positive for the majority of the 216 raw run-layer pairs, and once these are averaged within each seed to remove per-layer noise, the fitted slope is positive for all 54 of the resulting trajectories; this seed-averaged version is the stronger claim reported in the abstract. A raw run-layer pair can still show a negative $\text{PR}_Q - \text{PR}_K$ value in the earliest epochs, before the slope over the full training window has had a chance to establish itself, which is visible near the start of Figure 2. The interventions show that Key-side specialization is closely linked to attention sharpening and selectivity of attention. New checkpoint analyses on token-level models qualify the scope of this statement: the small-model trajectory is not a universal scaling law, but the $QK^\top$ interaction spectrum remains a useful bridge between key geometry and attention sharpness.

2. Related Work

2.1. Transformer Initialization and Training Stability

Earlier studies on Transformer initialization focused on the engineering aspect of making it an efficient factor for achieving better convergence and loss stabilization. [Glorot and Bengio \(2010\)](#) established variance-preserving initialization for deep networks, which became the baseline Xavier scheme used in many Transformer implementations. [Xiong et al. \(2020\)](#) showed theoretically that pre-normalization layer placement removes the depth-dependent gradient explosion of post-normalization architectures, making initialization scale less critical for stability. [Liu et al. \(2020\)](#) identified residual branch amplification as a proximate cause of post-normalization instability and proposed Admin initialization to correct it. [Huang et al. \(2020\)](#) eliminated LayerNorm and warmup through depth-scaled initialization, while [Wang et al. \(2022\)](#) extended stability analysis to very deep Transformers through DeepNorm.

Most of this literature evaluates initialization through optimization stability, convergence speed, or final loss. A broader theoretical perspective views initialization as shaping training-time function-space dynamics ([Jacot et al., 2020](#)). Recent work also suggests that random initialization can induce stable, seed-specific biases that persist throughout training, including SeedPrint-style model identities and structural token-preference biases present at initialization [Tong et al. \(2026\)](#); [Li et al. \(2026\)](#).

2.2. Rank Collapse and Entropy Collapse

A parallel line of work studies geometric degeneration inside Transformers as a pathology to be corrected. [Dong et al. \(2023\)](#) proved that pure attention without residual connections can contract representational rank rapidly with depth. [Noci et al. \(2022\)](#) linked initialization-depth interactions to query-key gradient vanishing in residual Transformers. [Zhai et al. \(2023\)](#) studied attention entropy collapse and proposed spectral reparameterization to pre-

vent unstable concentration. Closest to spectral analyses of attention, [Bao et al. \(2024\)](#) showed that self-attention networks localize when the QK eigenspectrum concentrates, while [Hong and Lee \(2025\)](#) identified softmax variance sensitivity as a driver of attention entropy collapse and training instability.

These studies examine various manifestations of representational collapse, including reductions in rank, concentration of attention, and degeneration of the interaction spectrum.

2.3. Q/K Subspace Structure and Sparse Attention

The authors in [Pan et al. \(2024\)](#) analyzed $W_Q^\top W_K$ using singular value decomposition (SVD) in Vision Transformers and identified semantic modes associated with token grouping and contextualization. Previous analyses of attention in pretrained Transformer models reported structured functional roles for attention heads, including syntactic and positional heads, as well as specialized heads that make disproportionately large contributions to model performance ([Clark et al., 2019](#); [Voita et al., 2019](#)). Other work demonstrated that attention weights alone are insufficient to explain attention outputs because the transformed value vectors and their norms also influence each token’s contribution to the resulting representation ([Kobayashi et al., 2020](#)). [Ataee Tarzanagh et al. \(2023\)](#) theoretically showed that gradient descent drives the attention softmax toward max-margin token selection. Finally, normalization of Q/K logits directly manipulates the attention-logit geometry, mitigating softmax saturation ([Henry et al., 2020](#)). Prior work often analyzes the combined query-key interaction object. Transformer circuit analyses likewise emphasize that the functional QK object is an operator rather than two isolated matrices [Elhage et al. \(2021\)](#). Induction-head analyses further illustrate how particular attention circuits emerge during training and support sequence-level computations ([Olsson et al., 2022](#)). Recent frozen-attention studies provide an additional perspective: [Dong et al. \(2025\)](#) showed that Transformers with frozen query and key weights can still form induction heads and remain competitive on language modeling, while MixiT isolates the role of random fixed attention.

3. Experimental Setup

3.1. Model

We use decoder-only GPT-style Transformers based on the NanoGPT architecture. Models use 4 attention heads per layer, hidden dimension $d_{model} = 256$, head dimension $d_{head} = 64$, MLP hidden dimension 1024, and context length 128. We evaluate 4-, 6-, and 8-layer variants, yielding approximately 4M–8M parameters depending on depth. All linear projections include bias terms, GELU activations are used in the feed-forward network, and dropout is disabled.

3.2. Dataset

We conduct our experiments on the character-level WikiText-103 corpus. We use the first 40 million characters of the corpus, split into 90 % training and 10 % validation sets. This produces approximately 36M training tokens, 4M validation tokens, and a vocabulary size of 1758. Character-level tokenization was chosen to preserve fine-grained sequence statistics and to make early attention dynamics visible in a computationally tractable setting.

3.3. Training Procedure

All models were trained using AdamW with learning rate 1×10^{-4} , $\beta_1 = 0.9$, $\beta_2 = 0.95$, and weight decay 0.1. Gradient clipping was applied at maximum norm 1.0. We used a constant learning rate schedule, batch size 64, sequence length 128, and mixed precision training with PyTorch AMP.

The baseline observational grid contains 3 depths, 3 initialization regimes, and 4 random seeds (37, 42, 105, and 210), for 36 baseline runs. Across all depths, initialization regimes, and seeds, the observational suite yields 216 run-layer trajectories over a shared 45-epoch analysis window. Unless otherwise stated, baseline statistics are computed over this 36-run grid; when independence is a concern, we report seed-averaged initialization-depth-layer traces rather than treating every layer trajectory as an independent model.

3.4. Initialization Regimes

We compare three query-key initialization strategies that explicitly control the geometric relationship between W_Q and W_K .

3.4.1. INDEPENDENT XAVIER

W_Q and W_K are initialized independently using the standard PyTorch Xavier-uniform initializer, producing near-zero expected alignment between query and key subspaces.

3.4.2. QK IDENTICAL

W_Q is initialized using Xavier initialization, and W_K is initialized as an exact copy of W_Q , producing perfect initial alignment between query and key projections.

3.4.3. DISJOINT SUPPORT XAVIER

W_Q and W_K are initialized with non-overlapping support. W_Q contains Xavier-initialized values in the first 128 columns and zeros elsewhere, while W_K contains zeros in the first 128 columns and Xavier-initialized values in the remaining 128 columns. This construction enforces zero initial overlap between active query and key subspaces.

3.5. Metrics

To characterize how initialization geometry influences optimization and representation formation, we track metrics spanning optimization behavior, attention concentration, query-key alignment, spectral structure, and representational scaling. Table 1 lists the tracked quantities. Unless otherwise stated, all metrics are computed layerwise throughout training and aggregated across random seeds.

The primary geometric metrics used throughout the study are participation ratio (PR) and effective rank:

$$\text{PR}(X) = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}. \quad (1)$$

$$\text{erank}(X) = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}. \quad (2)$$

Category	Metrics
Optimization	Training loss, validation loss
Attention Distribution	Attention entropy, top-1/top-5 attention mass, attention gap $p_{(1)} - p_{(2)}$
Query-Key Geometry	Cosine similarity between W_Q and W_K , column-wise alignment
Spectral Structure	Participation ratio (PR), effective rank, covariance eigenvalue statistics
Logit Geometry	Mean, variance, maximum attention logits
Norm and Scale	Activation norms, Frobenius norms, spectral norms

Table 1: Tracked geometric and optimization metrics.

Here λ_i are covariance eigenvalues and σ_i are singular values. In this paper, PR_Q and PR_K are computed from the covariance spectrum of the projected query and key activations for a layer after concatenating heads, so they measure layer-level effective dimensionality rather than per-head semantic feature count. Head-level quantities are used for $QK^\top$ effective rank and attention effective rank, then averaged across heads for the layer summary. Weight-spectrum diagnostics such as W_K SVD effective rank are computed directly from projection weights and are reported separately from activation PR and $QK^\top$ interaction-rank metrics.

3.6. Intervention Experiments

To test whether query-key geometry plays a causal role in attention organization and spectral evolution, we conducted a suite of controlled interventions on the 8-layer Transformer configuration. These experiments actively modified the dynamics of query and key projections during training, as summarized in Table 2.

Intervention	Manipulation
K Spectrum Clamp	Restores the singular-value spectrum of W_K to its initialization spectrum every 100 optimizer steps from epoch 1 onward
Forced K Contraction	Shrinks the W_K spectral tail after rank 32 by a factor of 0.05 every 100 optimizer steps, then renormalizes the projection
Freeze Q	Restores W_Q to its initialization reference every optimizer step
Freeze K	Restores W_K to its initialization reference every optimizer step
QK Decouple	Begins from equal Q/K initialization, then rotates W_K once after epoch 3 begins
QK Align Regularized	Adds a weak Q/K alignment regularization penalty with $\lambda = 0.001$

Table 2: Controlled interventions.

Each intervention was evaluated across five random seeds (7, 37, 42, 105, and 210) for 15 epochs using the same geometric metrics and logging framework as the baseline experiments.

3.7. Token-Level Checkpoint Extension

To test whether the small character-level results extend to larger token-level language models, we also measured activation-based query/key participation ratios and attention

geometry on public intermediate checkpoints. This extension covers GPT-2 small, Pythia 70M/160M/410M, SmolLM2 135M/360M, and OLMo-1B, spanning approximately 70M to 1B parameters. These models use their native tokenizers, corpora, checkpoint schedules, and standard independent query/key initialization; thus the extension is a boundary analysis rather than a controlled replication of the initialization grid. Six checkpoint series contain at least three revisions and support training-progress slope fits; OLMo-1B contributes only an endpoint layer-profile check.

4. Results

All baseline results are reported across 36 runs spanning four seeds, three depths, and three initialization regimes. Unless otherwise stated, trajectory statistics use seed-averaged initialization-depth-layer traces over the full 45-epoch baseline training horizon, while final geometry and loss summaries average over the corresponding trained runs. Validation losses remain close across initialization regimes: QK Identical reaches 1.1907 ± 0.0106 , Disjoint Support Xavier reaches 1.1969 ± 0.0112 , and Independent Xavier reaches 1.1973 ± 0.0108 . Pairwise Welch tests do not show a large loss separation.¹ We therefore treat the main effects below as differences in internal attention geometry rather than as simple consequences of different task loss.

4.1. Persistent Geometry and Layer Roles

The strongest persistent initialization signal is the final alignment between W_Q and W_K . QK Identical begins with alignment 1.0000, while Disjoint Support Xavier begins near 0 and Independent Xavier begins near 0.0498. After 45 epochs, the regimes remain separated in the aggregated final summaries: final alignment is 0.4530 ± 0.0950 for QK Identical, 0.1991 ± 0.0151 for Disjoint Support Xavier, and 0.1895 ± 0.0227 for Independent Xavier. The QK Identical versus uncoupled final-alignment contrast remains large at convergence.² Thus the initial angular relation is not erased by training. Figure 1 and Table 3 summarize this persistent separation.

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1. Pairwise Welch tests: QK Identical vs. Independent Xavier, $p = 0.144$, Hedges' $g = -0.598$; QK Identical vs. Disjoint Support Xavier, $p = 0.180$, $g = -0.545$; Independent Xavier vs. Disjoint Support Xavier, $p = 0.921$, $g = 0.039$.
 2. Welch test comparing QK Identical against the combined uncoupled regimes: $p = 1.06 \times 10^{-6}$, Hedges' $g = 4.489$.

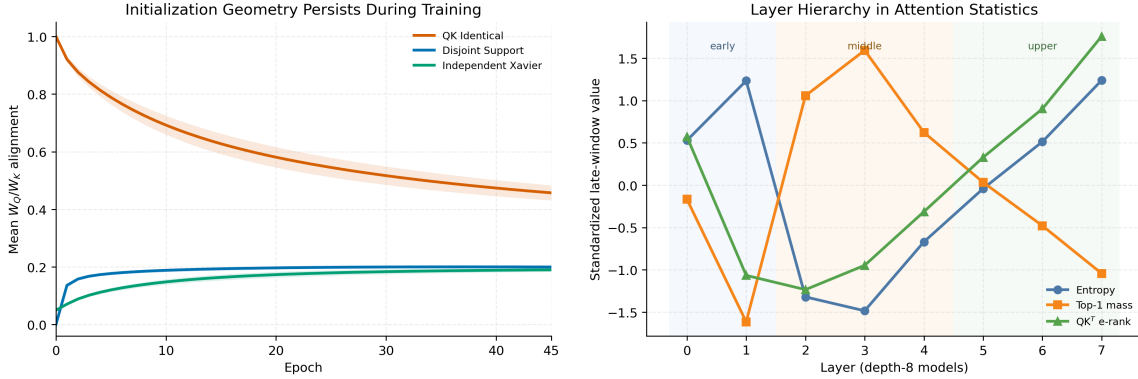


Figure 1: Initialization geometry and depth-wise attention roles. Top: W_Q/W_K alignment remains separated by initialization; bottom: depth-8 profiles reveal early, middle, and upper regimes.

Table 3: Final loss and alignment summary.

Initialization	Runs	Final val loss	Final alignment	Depth-8 alignment
QK Identical	12	1.1907 ± 0.0106	0.4530 ± 0.0950	0.5048 ± 0.0861
Disjoint Support Xavier	12	1.1969 ± 0.0112	0.1991 ± 0.0151	0.1911 ± 0.0077
Independent Xavier	12	1.1973 ± 0.0108	0.1895 ± 0.0227	0.1799 ± 0.0241

The persistence is also depth-dependent. At four layers, the final alignment gap between QK Identical and the strongest uncoupled regime is approximately 0.143; at eight layers, the same gap grows to approximately 0.314. Thus depth does not wash out the initialization signature. It amplifies the distance between the coupled and uncoupled geometric regimes while leaving final validation loss comparatively close.

The resulting aggregate final representations can be categorized into three different layer categories. Initial layers have higher entropy and QK^T effective rank, which is in line with keeping the token-level information intact. Mid layers make up the most constrained attention bottleneck with average entropy 1.532, average top-1 mass 0.514, and average QK^T effective rank 6.26. Higher layers increase the interaction space again as their QK^T effective rank increases to 16.34, while their PR_Q becomes 24.41, contrasted with the middle layer’s average value of 10.52. This confirms the outline-level description of the model training process.

4.2. Q-Expansion and Key Specialization

Expansion on the query side grows significantly in the seed-averaged trajectories. This trend is positive for nearly all seed-averaged initialization-depth-layer trajectories and exists in both uncoupled and coupled conditions. As a result, query expansion becomes an established aggregation trend rather than something that is specific to one particular seed or initialization depth.

Participation on the key side demonstrates a complementary specialization trend. Again, in the seed-averaged summary statistics, the slope of PR_K is negative for most trajectories and most strongly negative for uncoupled initialization. This means that key-side contraction is markedly weaker under the coupled condition, and reverses sign for a sizable minority of individual trajectories, consistent with the coupled initialization pulling key participation closer to the rising query trajectory. Table 4 reports these slope contrasts, and Figure 2 shows the corresponding trajectories.

Table 4: Participation-ratio slope summary.

Subset	n	PR_Q slope	PR_K slope	$(PR_Q - PR_K)$ slope
All seed-averaged traces	54	$+0.1406 \pm 0.0902$	-0.0437 ± 0.0565	$+0.1843 \pm 0.0966$
Uncoupled Q/K	36	$+0.1366 \pm 0.0932$	-0.0539 ± 0.0547	$+0.1904 \pm 0.0930$
QK Identical	18	$+0.1487 \pm 0.0859$	-0.0235 ± 0.0561	$+0.1722 \pm 0.1051$

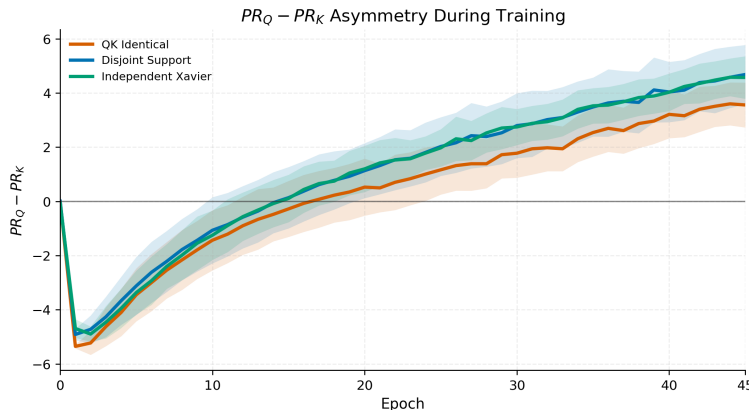


Figure 2: Query participation outpaces key participation. Curves show seed-wise averaged $PR_Q - PR_K$ trajectories; shaded bands show standard error across seeds.

The standard deviation on the pooled PR_K slope in Table 4 exceeds its mean, which reflects real heterogeneity in effect size rather than measurement noise: key contraction is strong and consistent under both uncoupled regimes, while under QK Identical it is markedly weaker and occasionally reverses, since the coupled initialization lets the rising query trajectory pull key participation upward at some depths instead of letting it contract. Pooling these regimes into a single mean and standard deviation is therefore expected to look noisy on PR_K alone, even though the direction of the effect still holds for the large majority of trajectories: key engagement decreases in 50 of the 54 seed-averaged traces overall.

Paired contrast is the most robust measure precisely because it is far less sensitive to this regime-dependent heterogeneity. The slope of $PR_Q - PR_K$ is positive for every seed averaged initialization depth layer path, whether uncoupled or coupled. It follows that the fundamental asymmetry is neither an expansion of the queries nor a contraction of the keys in isolation, but rather an increasingly broad query path compared to the key path.

Figure 3 summarizes the measured bridge from key specialization to interaction compression and entropy sharpening.

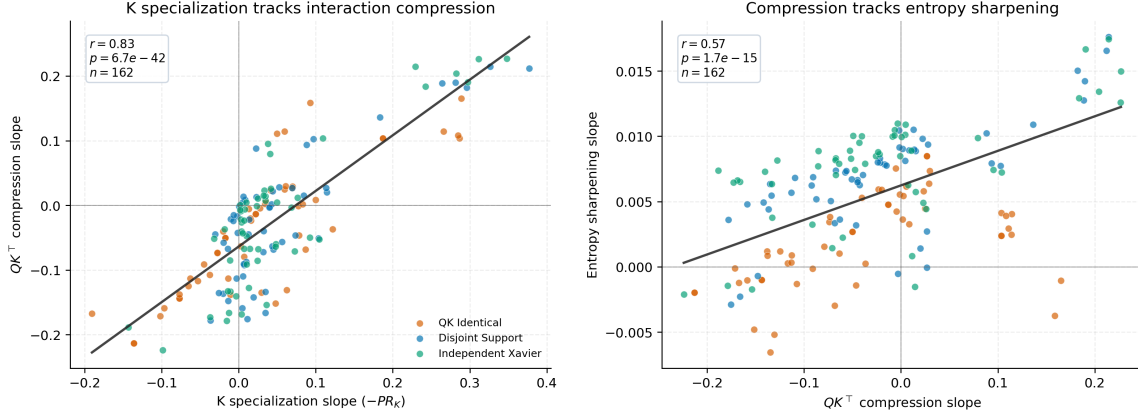


Figure 3: Key specialization predicts interaction-spectrum compression. Compression, in turn, predicts entropy sharpening.

This relationship is important since participation processes are correlated with both the level of attention sharpness and the structure of interactions-rank. Among the aggregate uncoupled run-layer dynamics, higher levels of key specialization are associated with higher relative compressions of the $QK^\top$ interaction spectrum ($r = 0.781$, $p = 8.32 \times 10^{-31}$, $n = 144$), which are in turn associated with entropy sharpening ($r = 0.614$, $p = 2.73 \times 10^{-16}$) and top-1 sharpening ($r = 0.488$, $p = 5.43 \times 10^{-10}$). These $n = 144$ observations pool all four seeds as separate points for the same eighteen layer-by-depth trajectories under each of the two uncoupled initializations, so the associated p -values are best read as descriptive rather than as independent-sample significance tests; the properly seed-averaged version of this same contrast, at the $n = 36$ resolution already used for the uncoupled subset in Table 4, is the analysis that respects the seed-independence treatment adopted elsewhere in this section. Query expansion is therefore best interpreted as a robust background adaptation, while key specialization is the more direct bridge to attention sharpening, with the seed-averaged paired trajectories in Table 4 providing the independence-respecting confirmation of the underlying asymmetry.

Log-only mechanism checks support the same interpretation after accounting for validation loss. In a loss-controlled epoch-layer regression, $QK^\top$ effective rank remains positively associated with mean attention entropy ($\beta = 1.610$, $p = 2.66 \times 10^{-176}$) and negatively associated with top-1 mass ($\beta = -1.357$, $p = 1.96 \times 10^{-117}$) after including PR_Q , PR_K , and validation loss as covariates. Temporal cross-lag tests are directionally consistent but weaker: early key specialization over epochs 0–10 predicts later entropy sharpening over epochs 10–45 in uncoupled runs ($r = 0.715$, $p = 3.62 \times 10^{-18}$), but early entropy sharpening itself remains an important predictor. Therefore these checks are treated as robustness evidence, not as a replacement for intervention experiments.

4.3. Controlled Interventions on K Geometry

The controlled interventions test whether K-side geometry is merely correlated with attention behavior or can directly modulate it. The observational mechanism predicts that forcing K contraction should sharpen attention by reducing entropy and increasing logit scale, while preserving or freezing the K spectrum should soften attention by maintaining a broader key subspace. The seed-aggregated intervention outcomes in Table 5 match this direction: forced K contraction has lower entropy and higher logit standard deviation than the K-preserving interventions, while K-spectrum clamp and freeze K preserve broader interaction spectra.

Table 5: Seed-aggregated intervention outcomes across seeds 7, 37, 42, 105, and 210. Confidence is the fraction of seed-metric comparisons whose final-epoch delta has the same sign as the aggregate delta for entropy, logit standard deviation, and $QK^\top$ effective rank.

Mode	Val loss	Entropy	Logit std	$QK^\top$ erank	Conf.
No intervention	1.2066	2.1893	4.4329	9.71	—
K spectrum clamp	1.2159	2.2888	3.5898	11.13	1.00
Forced K contraction	1.2270	2.0809	5.4071	6.03	1.00
Freeze Q	1.2196	2.5532	3.5915	9.35	0.93
Freeze K	1.2172	2.6002	3.2136	10.83	1.00
Q/K decouple	1.2115	1.7432	5.0723	9.92	0.93
Q/K alignment regularized	1.2153	2.1767	4.6830	9.50	0.73

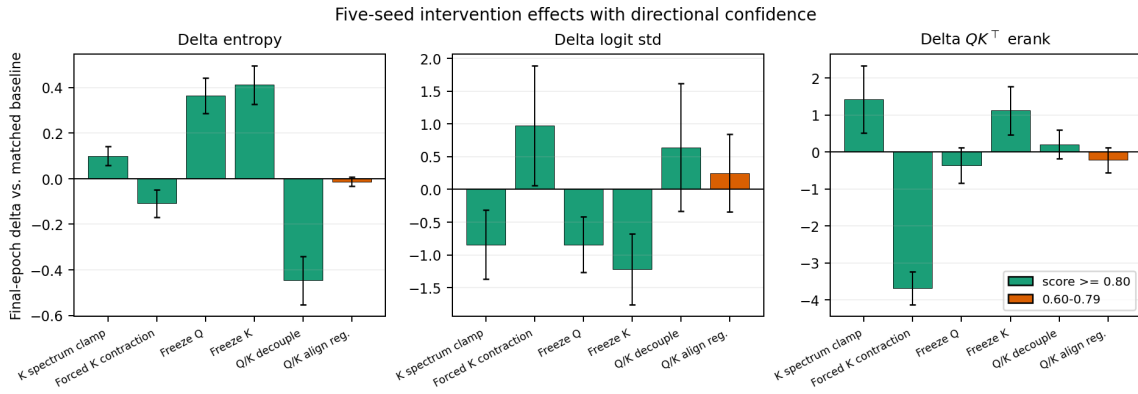


Figure 4: Five-seed intervention deltas against each seed’s matched no-intervention baseline. Bars show final-epoch mean deltas, error bars show 95% confidence intervals across seeds, and color indicates directional confidence score.

The seed-aggregated intervention results show the same qualitative direction across all five seeds. Forced K contraction yields the lowest mean entropy among the independent-initialized intervention conditions (2.0809), the largest mean logit standard deviation (5.4071), and the lowest $QK^\top$ effective rank (6.03), with confidence score 1.00. In contrast, freezing K gives mean entropy 2.6002, mean logit standard deviation 3.2136, and confidence score 1.00 in the opposite softening direction. K-spectrum clamp also softens attention relative to the matched no-intervention baseline, with higher entropy, lower logit scale, larger $QK^\top$ effective rank, and confidence score 1.00. These effects occur with aggregate final validation losses still in a narrow range from 1.2066 to 1.2270 for the independent-initialized intervention family.

The intervention suite also separates K specialization from adjacent explanations. Freezing Q softens attention even though it does not directly freeze the key pathway, indicating that query-side dynamics still affect the final attention state. However, the K-specific manipulations produce the clearest monotonic pattern: contraction sharpens, while preservation or freezing softens. The weak Q/K alignment regularizer produces only modest aggregate changes relative to the stronger K-specific manipulations, which argues against an alignment-only account. These interventions support the interpretation that K geometry acts as a causal control surface for the $QK^\top$ interaction spectrum and, through it, attention entropy and logit scale.

4.4. Scaling Behavior on Public Checkpoints

The GPT-2 small trajectory is the most informative point in this extension, since it is the only checkpoint series with a dense, full-training arc rather than a handful of widely spaced snapshots. It shows an early query-over-key participation burst rather than a persistent monotonic increase: the model-average $PR_Q - PR_K$ contrast rises sharply within the first several percent of training and then decays as key participation recovers, so the fitted full-trajectory slope is negative (-0.1098 after normalization by head dimension) and no GPT-2 layer has a positive fitted contrast slope. Read together with the controlled character-level grid, which is trained for a comparatively short, fixed horizon, this suggests that the small-model result characterizes the same early regime that GPT-2 small passes through and later leaves behind, rather than a property expected to hold for the full duration of pretraining.

Table 6 extends the comparison across five further checkpoint series – Pythia 70M, 160M, and 410M, and SmolLM2 135M and 360M – each with at least three revisions, plus an endpoint-only check on OLMo-1B. Across the six series with fitted slopes, the normalized $PR_Q - PR_K$ slope is positive in only one model (Pythia-70M) and in 17 of 116 model-layer trajectories, so the raw participation-ratio slope observed in the controlled grid does not carry over as a general cross-family scaling law once training extends well beyond the early regime described above.

Table 6: Token-level checkpoint extension. Counts use only checkpoint series with at least three revisions. Bridge co-movement indicates positive correlation between PR_K and $QK^\top$ effective rank together with positive correlation between $QK^\top$ effective rank and attention entropy.

Family	Models	ΔPR slope > 0	PR_Q slope > 0	PR_K slope < 0	Bridge co-move
Pythia	3	1/3	0/3	3/3	3/3
SmolLM2	2	0/2	0/2	0/2	0/2
GPT-2	1	0/1	1/1	0/1	1/1
All fitted series	6	1/6	1/6	3/6	4/6

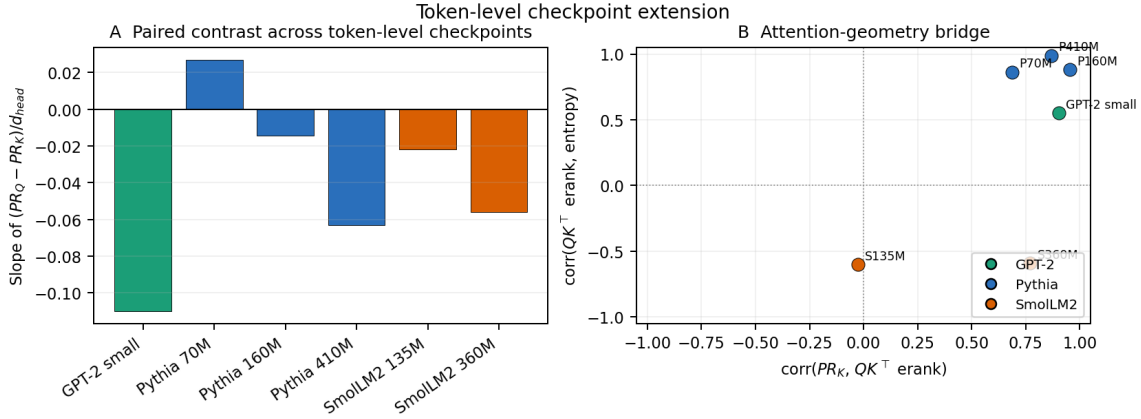


Figure 5: Token-level checkpoint extension. Left: fitted slopes of the normalized paired contrast $(\text{PR}_Q - \text{PR}_K)/d_{\text{head}}$ across public checkpoint series. Right: model-level correlations connecting key participation, $QK^\top$ effective rank, and attention entropy.

The three Pythia models are the clearest case of this early-regime signature persisting further into training: key participation contracts in all three, and each shows the same directional co-movement between PR_K , $QK^\top$ effective rank, and attention entropy that GPT-2 small exhibits ($r = 0.905$ for PR_K vs. $QK^\top$ effective rank and $r = 0.551$ for $QK^\top$ effective rank vs. entropy in GPT-2 small; see Table 6 for the corresponding co-movement counts). This PR_K -to- $QK^\top$ -effective-rank-to-entropy bridge is therefore more portable across families than the raw participation-ratio slope, holding in GPT-2 small and all three Pythia models. It is not universal, however: neither SmolLM2 series shows key contraction or the same bridge co-movement, so the bridge is best read as a frequently-recurring regularity rather than a scale-invariant law.

5. Discussion

This work aimed to answer the question, how does the effective dimensionality of query and key representations evolve during the training run, and how does initialization affect it. After running the diagnostic experiments, it was observed that the PR_Q increases throughout the training run, while PR_K decreases. The dynamics of participation ratio might be due to complex interaction of initialization, gradients and architecture. We observe that initialization does affect the trajectory of the ratios.

The increase in PR_Q suggests that queries come to occupy a broader activation subspace over training, while the decrease in PR_K means keys settle into a narrower one. This is a statement about the spread of the projected query and key activations, not about the rank of the weight matrices W_q and W_k themselves; we track weight-level SVD effective rank separately, and it is not the quantity behind this trend. A broader query space may allow tokens to interact with a larger number of key projections, potentially making the attention more uniform. The contracting key space is consistent with lower entropy from the intervention. Both phenomenon, suggesting that effective attention comes out as a balance between the PR_Q and PR_K .

Beyond giving an insight into the training process of Transformers, the above-mentioned findings indicate that we can use geometric values like PR_Q and PR_K for diagnostic purposes to check if the attention is uniformly distributed due to lack of specialization in keys. Such an approach will help in determining the reason behind some unwanted behavior of attention and suitable changes can be made accordingly.

The larger checkpoint extension helps situate this diagnostic within the arc of a full pretraining run. In GPT-2 small, the query-over-key participation gap rises sharply in the first several percent of training and then decays as key participation recovers; the controlled character-level experiments, trained for a comparatively short and fixed horizon, appear to be characterizing this same early regime rather than a property that holds across the whole of pretraining. Under this reading, query expansion and key specialization are best understood as an early-training signature of how attention geometry initially organizes itself, rather than a monotonic law that must hold at every stage of training or at every scale. The PR_K -to- $QK^\top$ -effective-rank-to-entropy bridge discussed above is a comparatively more portable signal across the checkpoint families tested, though it is likewise not a scale-invariant law.

6. Limitations

The main experiments are performed on small character-level GPT-like decoder transformers trained on WikiText-103 with context length 128 to permit dense geometric logging with restricted computational resources. The token-level checkpoint extension partially addresses external validity, but it should be interpreted as a boundary analysis rather than a controlled replication: the public checkpoint families differ in tokenizer, corpus, architecture, checkpoint spacing, and training recipe, and they do not permit the same query-key initialization manipulations. Larger controlled token-level experiments, retrieval-focused tasks or longer context length settings, other datasets, longer training schedules, and even transformer variants with grouped/multi-query attention would clarify the extent to which these dynamics persist across different Transformer architectures and variants. In particular,

because the GPT-2 checkpoint series shows the query-over-key asymmetry peaking early and then decaying, an open question is whether extending the controlled character-level training horizon well past the current 45 epochs would reveal a similar late-training reversal within the initialization grid itself. As noted above, the interaction-rank-to-entropy bridge is more portable than the raw participation-ratio slope but is not itself scale-invariant, so it should not be over-relied upon as a fallback universal claim.

The participation ratio is a geometric metric of spread or effective dimensionality, not of richness in semantics. This means that the increases in PR_Q reflect usage of more spread effective subspace, not richer semantics. Because $\text{PR}(X)$ is invariant to uniform rescaling of the activations, a simple growth in overall activation norm cannot by itself explain the reported PR_Q/PR_K trends; however, we have not separately verified that direction-dependent (anisotropic) norm growth is not contributing to the measured spectral shifts, and cleanly isolating this from genuine angular reorganization of the query/key subspaces is left as an open question for future work.

Observational evidence is provided for most baselines, however, consistency of the slopes comparison and correlation with cross-metric values provides the clear empirical structure throughout training schedule. Intervention experiments provide additional support for this interpretation and are now balanced across five seeds for the reported 8-layer setting, but they remain short-horizon mechanistic probes. Further ablation over the frequency of interventions, strength of clamp, threshold for ranks, head-wise vs. layer-wise intervention, interactions on the optimizer side are expected. The GPT-2 small extension also combines an early smoke run with a fuller checkpoint sweep, so a clean remeasurement with uniform token budgets would be preferable for stronger scaling claims.

7. Conclusion

This work sought to explore whether query and key projections undergo parallel development across initialization, depth, and layers during Transformer training. From 36 baseline runs, query engagement grew in 49 out of 54 seed-based trajectories, while key engagement decreased in 50 out of 54, and the difference $\text{PR}_Q - \text{PR}_K$ increased in all 54.

The findings also highlight the importance of the discussed asymmetry. It appears that key specialization is more strongly correlated with attention sharpening than query engagement, and that it is connected with the geometry of projections and relative compression of the $QK^\top$ interaction spectrum. The performed interventions support the causal sensitivity of this pathway: forced compression of the K spectrum reduced entropy and increased logit scale with confidence score 1.00, whereas maintaining or freezing K spectrum made attention softer with confidence score 1.00. Token-level checkpoint analyses qualify the scope of the result: monotonic growth of $\text{PR}_Q - \text{PR}_K$ is not universal across pretrained families, but the link between interaction-rank structure and attention entropy remains the more stable cross-model signal.

The general conclusion from this research is that initialization is not just an optimization starting point but rather a geometric prior on the query-key interface. Attention can be constructed using different internal geometries despite reaching identical validation losses.

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